

Research Interest

Cosmological simulations; dwarf galaxies; ultra-faint dwarfs; machine learning; disk formation; direct N-body methods; star cluster formation

Education

Seoul National University, Seoul

Mar. 2017 – Feb. 2021

Bachelor of Science - Double Major in Physics Education and Astronomy

Seoul National University, Seoul

Mar. 2021 – Present

MS/PhD Student in Physics

Academic Positions

Flatiron Institute, New York, NY

Jan. 2026 – May. 2026

Guest Researcher / Pre-Doctoral Researcher, Center for Computational Astrophysics (CCA)

Primary Refereed Publications

The AGORA High-resolution Galaxy Simulations Comparison Project. VIII: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$. *ApJ*. 994, 245 (2025)

M. Jung^{*}, J. -H Kim^{*}, T. H. Nguyen^{*}, R. Rodríguez-Cardoso^{*}, and other 21 co-authors for the AGORA Collaboration

Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions. *ApJ*. 965, 156 (2024)

M. Jung^{*}, Kim, J. -H^{*}, B. K. Oh, S. E. Hong, J. Lee, and J. Kim

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo. *ApJ*. 964, 123 (2024)

M. Jung^{*}, S. Roca-Fàbrega^{*}, J. -H Kim^{*}, and other 18 co-authors for the AGORA Collaboration

The AGORA High-resolution Galaxy Simulations Comparison Project. X: Formation and Evolution of Galaxies at the High-redshift Frontier. *Accepted to ApJ, arXiv:2511.04435*

H. Kim^{*}, J. -H Kim^{*}, **M. Jung**^{*}, J. -H Kim^{*}, S. Roca-Fàbrega^{*}, and other 20 co-authors for the AGORA Collaboration

Evolution of Nuclear Star Cluster in Dwarf Galaxy through Mergers and In-Situ Star Formation. *Accepted to ApJ, arXiv:2603.18242*

Y. Jo^{*}, **M. Jung**^{*}, S. Kim, G. L. Bryan, J. -H Kim, and Lee, A.

The AGORA High-resolution Galaxy Simulations Comparison Project. VII: Satellite quenching in zoom-in simulation of a Milky Way-mass halo. *A&A*. 698, A303

R. Rodríguez-Cardoso^{*}, S. Roca-Fàbrega^{*}, **M. Jung**^{*}, T. H. Nguyen^{*}, and other 18 co-authors for the AGORA Collaboration

The AGORA High-resolution Galaxy Simulations Comparison Project. IV: Halo and Galaxy Mass Assembly in a Cosmological Zoom-in Simulation at $z \leq 2$. *ApJ*. 968, 125 (2024)

S. Roca-Fàbrega^{*}, J. -H Kim^{*}, J. R. Primack^{*}, **M. Jung**^{*}, and other 23 co-authors for the AGORA Collaboration

Dark Matter Deficient Galaxies Produced Via High-velocity Galaxy Collisions In High-resolution Numerical Simulations. *ApJ*. 899, 25 (2020).

E. -J. Shin[†], **M. Jung**[†], G. Kwon[†], J. -H Kim^{*}, J. Lee, Y. Jo, B. K. Oh

Other Refereed Publications

Inferring Cosmological Parameters on SDSS via Domain-generalized Neural Networks and Light-cone Simulations. *ApJ*. 975, 38 (2024)

J. -Y Lee, J. -H Kim^{*}, **M. Jung**^{*}, and other 6 co-authors

The AGORA High-resolution Galaxy Simulations Comparison Project. VI: Similarities and Differences in the Circumgalactic Medium. *ApJ*. 962, 29 (2024)

S. Strawn^{*}, S. Roca-Fàbrega^{*}, J. R. Primack^{*}, J. -H Kim^{*}, and other 28 co-authors including **M. Jung**

^{*}corresponding author

[†]these authors contributed equally to this work

Awards

Outstanding Presentation Award

Oral Session (#)

Apr. 23, 2023

The Korean Physical Society

Outstanding Project Award (1st Place)

The 14th KIAS CAC Summer School on Artificial Intelligence & Parallel Computing (#)

June. 30, 2023

Korea Institute for Advanced Study

Outstanding Project Award (3rd Place)

2024 Winter School on Programming for Accelerators - Advanced Course (#)

Feb. 24, 2024

SNU THUNDER Research Group

Oral Presentations

Numerical Galaxy Formation Mini-Workshop (#)

Jan. 16, 2020

Dark matter deficient galaxies produced via high-velocity galaxy collisions in high-resolution numerical simulations

Shin, E. -J & M. Jung

2023 Darwin+NGF Workshop (#)

Jan. 11, 2023

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo

M. Jung

2023 KPS spring meeting (#)

Apr. 21, 2023

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo

M. Jung

XV International Conference on Gravitation, Astrophysics and Cosmology (ICGAC15, #)

Jul. 3, 2023

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo

M. Jung

2023 Santa Cruz Galaxy Workshop (#)

Aug. 10, 2023

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo

M. Jung

SNU ARC 2nd H.S. Yun Astronomy Workshop (#)

Aug. 30, 2023

The AGORA High-resolution Galaxy Simulations Comparison Project. V: Satellite Galaxy Populations In A Cosmological Zoom-in Simulation of A Milky Way-mass Halo

M. Jung

2023 108th KAS Fall Meeting (#)

Oct. 18, 2023

Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions

M. Jung

The 2nd Workshop on Galaxies and Dark Matter (Hosted by SNU LAMP Foundation)

Feb. 27, 2024

The AGORA High-resolution Galaxy Simulations Comparison Project:

A MW-mass Galaxy and Substructures in a Cosmological Zoom-in Simulation

M. Jung

The 2nd CTP Bosan Workshop: AGORA in Asia + 5th Numerical Galaxy Formation Meeting in Korea (#)

May. 7, 2024

The Satellite Galaxy Population in the AGORA CosmoRun, and Resolution Convergence Test in the TNG Simulation

M. Jung

2024 110th KAS Fall Meeting (#)

Oct. 16, 2024

The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$

M. Jung

The 3rd Workshop on Galaxies and Dark Matter (#)

Feb. 27, 2025

The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$

M. Jung

2025 Santa Cruz Galaxy Workshop (#)

Aug. 8, 2025

The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$

M. Jung

SNU ARC 4th H.S. Yun Astronomy Workshop (#)

Aug. 21, 2025

The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$

M. Jung

CCA/NY Area Galaxy Formation Seminar

Jan. 30, 2026

Star Clusters and Galaxies Using Self-consistent Hybrid Hydro/direct N-body Simulations

M. Jung

2026 CCA Pre-Doc Symposium (#)

May. 19, 2026

Fate of ram-pressure stripped gas from resolved dwarf galaxies

M. Jung

1st Galaxy Workshop and AGORA in Europe (#)

May. 27, 2026

Fate of ram-pressure stripped gas from resolved dwarf galaxies

M. Jung

Poster Presentations

2023 DARWIN-Dwarf galaxy researcher workshop (#)

Aug. 16, 2023

Merger-tree-based Galaxy Matching: A Comparative Study across Different Resolutions

M. Jung

The 11th KIAS Workshop on Cosmology & Structure Formation (#)

Oct. 27, 2024

The AGORA High-resolution Galaxy Simulations Comparison Project. IX: Formation and Evolution of Disk in Milky Way Mass Progenitor Galaxies at $1 < z < 5$

M. Jung

Experiences

Undergraduate Internship

Mar. 2019 – Aug. 2020

Department of Physics & Astronomy (Advisor: Prof. Kim Ji-hoon)

SNU

- Studied cosmological hydrodynamical simulations
- Investigated the kinematics of dark matter and baryons in galaxies via simulations
- Conducted a statistical analysis of the spin of galaxies

System Administrator

Sep. 2021 – Present

Computational Cosmology Group (Advisor: Prof. Kim Ji-hoon)

SNU

- Administered the lab's computing cluster with root access; managed user environments, installed scientific software, and ensured system stability

Teaching Experiences

Assistant Mentor, 1 year, SNU Science Education Institute for The Gifted, Student Research Mentoring Program (Physics)	2022
Teaching Assistant, 1 semester, Seoul National University, Rudimentary Mathematical Methods of Physics	2022
Teaching Assistant, 1 semester, Seoul National University, Physics 1	2022
Teaching Assistant, 1 semester, Seoul National University, Rudimentary Mathematical Methods of Physics	2021
Teaching Assistant, 1 semester, Seoul National University, Physics Lab.1	2021
Teaching Practicum, 1 semester, Seoul National University Middle School	2020

Collaborations and Technical Skills

Collaborations: AGORA, FOGGIE

Programming Languages: Python, C, C++, CUDA

Programs: ENZO, GADGET-4, MUSIC

Libraries: yt, scipy, pandas, pytorch

Open-Source Contributions: ABYSS, ENZO-ABYSS